Video anomaly detection on a 4 GB laptop GPU

Frozen vision-language encoder + two small trained heads · 11 classes · 20.3x real time on an RTX 3050 Laptop

60.7 / 100
ARENA SCORE · PUBLIC TEST

PIPELINE

1. **Drone / CCTV / dashcam video**
2. **Sample 2 fps downsize on decode** — OpenCV
3. **SigLIP-2 base FROZEN, fp16** — 78 fps · 1.03 GB
4. **4 s windows mean/max/std/drift** — 2 s hop
5. **Two MLP heads window + clip** — trained, <1 min
6. **Hysteresis decoder relative to baseline** — merge · gate

Encode once, reuse forever.
The encoder is frozen, so all 3,207 videos are embedded once into a cache. Retraining a head then takes under a minute, which is what made same-day iteration on windowing and thresholds possible.

WHY NOT RUN A VLM PER FRAME

4 GB rules out fine-tuning a 2B VLM locally. So we froze one and trained only what sits on top.

The always-on stage is, in the end, a 12-way decision — orders of magnitude cheaper as a head over frozen embeddings than as generative decoding. The open-vocabulary semantics still come from the VLM; only the task-specific part is learned.

Explanations use nearest-neighbour retrieval over training descriptions in the same embedding space — one dot product per event, no generative model.

SPEED, MEASURED END TO END

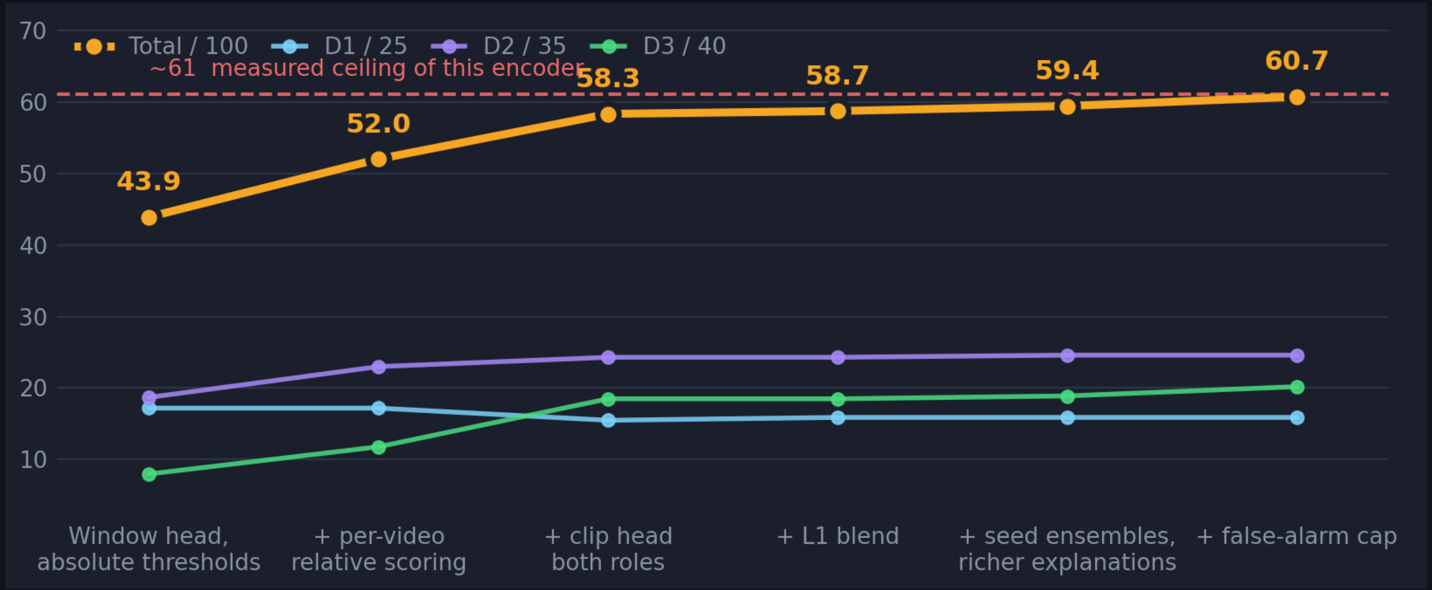
3,391 s
of video

167 s
to process, incl. decode

Decode is the bottleneck, not the GPU.
Encoding 2 fps costs 1.03 GB and leaves headroom for many streams.

What moved the score, and what didn't

Real arena marks across 15 scored runs. D1 is worth 25, D2 35, D3 40.



THREE THINGS THAT MATTERED

Score against each video's own baseline

The head, trained on short clips where the event fills the frame, reports a high flat pedestal on long footage — baselines ranged 0.55 to 0.88, so no absolute threshold transfers. Thresholding on the rise above each video's own median found the events.

L2 +0.12 · L3 +0.10

Use each head for its own question

The window head answers where; the clip head answers what. Letting the clip head classify each detected span, instead of pooling window votes, fixed whole videos at once.

L3 0.296 → 0.414

We proved when to stop tuning

Oracle search over ~15,000 decoder configs, scoring each video at its own best setting, tops out near 61 marks — we are at 60.7. Two D3 videos sit exactly at their ceiling: no threshold can rank their events.

Decoder is finished; encoder is the wall

WHAT WE TRIED AND DROPPED

Contrast-to-baseline features: +0.8% held-out, but test AUC collapsed 0.86→0.53 · logit adjustment for rare classes: zero lift · TTA: zero lift

FINAL, BY DIFFICULTY TIER

Difficulty 1	Difficulty 2	Difficulty 3
63.7%	**70.3%**	**50.4%**
15.9 / 25 marks	**24.6 / 35 marks**	**20.2 / 40 marks**
class confusion is the wall	oracle ceiling 24.8	oracle ceiling 19.8

Honest caveat.
The decoder is tuned on 6 Difficulty-2 and 4 Difficulty-3 videos, so those thresholds are the least trustworthy part of the system, and ties break toward the least aggressive setting. Our scorer matches the published structure of the metric; the exact within-tier weighting isn't published, so it ranks configurations rather than predicting the leaderboard.